# History of Python

## 1. Introduction

Python is a popular, high-level, general-purpose programming language.

Python was created by **Guido van Rossum**.

Python was designed to be:

- Easy to read
- Easy to learn
- Simple to write
- Powerful
- Flexible
- Useful for many types of applications

Python development started in the **late 1980s**, and the first public version was released in **1991**.

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# 2. Who Created Python?

Python was created by:

**Guido van Rossum**

Guido van Rossum is a Dutch programmer.

He started working on Python as a personal programming project.

He wanted to create a programming language that was:

- Simple
- Readable
- Practical
- Easy to use
- Powerful enough for real programming

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# 3. When Was Python Created?

Python development began around:

**1989**

The first public release of Python was:

**1991**

So, Python has been developed and improved for many years.

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# 4. Why Was Python Created?

Before Python, programmers commonly used languages such as:

- C
- C++
- Pascal
- ABC
- Perl
- Lisp

Many programming languages were powerful, but some were difficult for beginners to read and write.

Guido van Rossum wanted a language that could combine:

- Simplicity
- Readability
- Productivity
- Extensibility
- Powerful features

Python was created with these ideas in mind.

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# 5. ABC Programming Language

An important influence on Python was the **ABC programming language**.

Guido van Rossum had worked at **CWI** in the Netherlands.

ABC was designed to make programming easier, especially for beginners.

Python took some ideas from ABC but also improved and changed many things.

Python was designed to be more practical and extensible.

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# 6. Python Development Started

During the Christmas holidays of **1989**, Guido van Rossum started working on Python.

He worked on the project while at CWI in the Netherlands.

The project was originally created as a successor to the ABC programming language.

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# 7. Why Is It Called Python?

The name **Python** did not come from the snake.

Guido van Rossum was a fan of the British comedy group:

**Monty Python**

He was watching the comedy series:

**Monty Python's Flying Circus**

when he was working on the language.

He wanted a name that was:

- Short
- Unique
- Interesting
- Easy to remember

So he chose the name:

**Python**

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# 8. First Public Release

The first public version of Python was released in:

**1991**

This early version already included important features such as:

- Functions
- Classes
- Exception handling
- Modules
- Core data types
- Dynamic typing

Python was already designed to be a high-level and readable language.

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## 9. Python 0.9.0

The first public release was:

**Python 0.9.0**

Released in:

**1991**

It was released through the early Internet community.

Python 0.9.0 included many features that are still important concepts in modern Python.

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## 10. Python 1.0

Python **1.0** was released in:

**1994**

Python continued to grow and gain new users.

The language became more mature and useful for software development.

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## 11. Python 2.0

Python **2.0** was released in:

**2000**

Python 2 introduced several important improvements.

Some notable improvements included:

- List comprehensions
- Garbage collection improvements
- Unicode support
- Better language features
- Improved standard library

Python 2 became widely used.

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# 12. Python 2 Era

For many years, Python 2 was one of the most commonly used Python versions.

Many:

- Companies
- Developers
- Universities
- Projects
- Open-source communities

used Python 2.

However, Python 2 had limitations that eventually required a major improvement.

This led to Python 3.

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# 13. Python 3

Python **3.0** was released in:

**2008**

Python 3 was designed as a cleaner and improved version of Python.

It introduced changes that were not fully backward compatible with Python 2.

This means:

```
Python 2 code
      ↓
may not work directly
      ↓
Python 3
```

Developers often had to modify older Python 2 programs when moving to Python 3.

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# 14. Why Was Python 3 Created?

Python 3 was created to improve the language and remove some old design problems.

The goals included:

- Cleaner language design
- Better consistency
- Better Unicode support
- Improved syntax
- Better long-term development

Python 3 became the future direction of Python.

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# 15. Python 2 vs Python 3

A simple comparison:

| Feature | Python 2 | Python 3 |
|---|---|---|
| First major release | 2000 | 2008 |
| Modern development | No | Yes |
| `print` | Statement | Function |
| Unicode | More complicated | Better built-in support |
| Division behavior | Different | Improved |
| Current learning choice | No | Yes |

Example:

### Python 2

```
print "Hello"
```

### Python 3

```
print("Hello")
```

Today, new Python learners should learn **Python 3**.

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## 16. End of Python 2

Python 2 officially reached the end of its life on:

**January 1, 2020**

After this date, the Python core development community stopped providing official Python 2 maintenance and security updates.

Python 3 is now the standard version for modern Python development.

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## 17. Python Software Foundation

The **Python Software Foundation (PSF)** was created in:

**2001**

The PSF is a non-profit organization that supports the Python community.

It helps with:

- Python development
- Community activities
- Conferences
- Education
- Python-related projects
- Open-source ecosystem

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## 18. Python Becomes Open Source

Python is an **open-source** programming language.

This means its source code is available for people to:

- Study
- Use
- Modify
- Improve
- Share

Python's open-source nature helped it grow into a large global community.

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## 19. Python Community

Python has a very large worldwide community.

Millions of developers use Python for different purposes.

The community contributes:

- Libraries
- Frameworks
- Tools
- Documentation
- Tutorials
- Open-source projects

This makes Python useful beyond the core language itself.

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## 20. Python Standard Library

One reason Python became popular is its large **standard library**.

The standard library provides modules for many common tasks.

Examples include:

```
import math
```

```
import os
```

```
import datetime
```

```
import json
```

This allows programmers to perform many tasks without installing additional packages.

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# 21. Python and Web Development

Python became popular in web development.

Python web frameworks include:

- Django
- Flask
- FastAPI

Example:

```
from fastapi import FastAPI

app = FastAPI()
```

Python can be used to create:

- Websites
- APIs
- Backend systems
- Web services

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# 22. Python and Data Science

Python later became extremely popular in:

**Data Science**

Developers use Python for:

- Data analysis
- Data processing
- Statistics
- Visualization
- Scientific computing

Popular libraries include:

```
NumPy
Pandas
Matplotlib
SciPy
```

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# 23. Python and Artificial Intelligence

Python is also widely used in:

**Artificial Intelligence**

Python can be used for:

- Machine learning
- Deep learning
- Natural language processing
- Computer vision
- AI research

Popular libraries include:

```
Scikit-learn
TensorFlow
PyTorch
Keras
```

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# 24. Python and Automation

Python is commonly used for automation.

For example, Python can automate:

- File operations
- Data processing
- Web tasks
- Reports
- Repetitive tasks
- System administration

Simple example:

```
print("Automation with Python")
```

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## 25. Python and Education

Python is widely used for teaching programming.

Why?

Because Python syntax is relatively simple.

Example:

```
name = "Sachin"

print(name)
```

A beginner can understand this code easily.

Python is therefore commonly used in:

- Schools
- Colleges
- Universities
- Programming courses
- Coding bootcamps

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## 26. Python and Scientific Computing

Python is also used in scientific and engineering fields.

Examples include:

- Mathematics
- Physics
- Biology
- Chemistry
- Astronomy
- Engineering

Python provides many scientific libraries that help researchers process and analyze data.

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# 27. Python Today

Python has grown from a small programming project into one of the world's most widely used programming languages.

It is used in many areas:

```
Python
   │
   ├── Web Development
   ├── Data Science
   ├── Artificial Intelligence
   ├── Machine Learning
   ├── Automation
   ├── Scientific Computing
   ├── Education
   ├── Testing
   ├── Software Development
   └── Data Analysis
```

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# 28. Simple Python History Timeline

```
1989
 │
 └── Python development begins
      │
1991
 │
 └── Python 0.9.0 released
      │
1994
 │
 └── Python 1.0 released
      │
2000
 │
 └── Python 2.0 released
      │
2001
 │
 └── Python Software Foundation created
      │
2008
 │
 └── Python 3.0 released
      │
2020
 │
 └── Python 2 reaches end of life
      │
Today
 │
 └── Python 3 is used for modern development
```

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# 29. Important People

## Guido van Rossum

Guido van Rossum is the creator of Python.

He started the Python project in the Netherlands.

He remained an important figure in Python development for many years.

He was also known as:

**Benevolent Dictator For Life (BDFL)**

for the Python project until stepping down from that role in 2018.

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# 30. Python and Guido van Rossum

Simple relationship:

```
Guido van Rossum
       ↓
Created Python
       ↓
Python development
       ↓
Python community
       ↓
Python ecosystem
```

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# 31. Important Historical Milestones

| Year | Event |
|---|---|
| 1989 | Python development begins |
| 1991 | Python 0.9.0 released |
| 1994 | Python 1.0 released |
| 2000 | Python 2.0 released |
| 2001 | Python Software Foundation created |
| 2008 | Python 3.0 released |
| 2018 | Guido van Rossum steps down as BDFL |
| 2020 | Python 2 reaches end of life |

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# 32. Why Python Became Popular

Python became popular for several reasons.

## 1. Easy Syntax

Python code is generally easy to read.

Example:

```
print("Hello World")
```

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## 2. Large Standard Library

Python provides many useful modules.

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## 3. Large Community

Millions of developers use and contribute to Python.

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## 4. Many Libraries

There are libraries for:

- Web development
- Data science
- AI
- Machine learning
- Automation
- Scientific computing

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## 5. Cross-Platform

Python can run on operating systems such as:

- Windows
- Linux
- macOS

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## 6. Open Source

Python is freely available and open source.

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# 33. Python Evolution

Python has continuously evolved.

The basic idea has remained:

> Make programming easier and more productive.

Over time Python gained:

- Better syntax
- Better performance
- Better libraries
- Better tooling
- Better Unicode support
- Type hints
- Async programming
- Modern development features

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## 34. Python Versions

Python has gone through several major versions.

```
Python 0.x
   ↓
Python 1.x
   ↓
Python 2.x
   ↓
Python 3.x
```

The most important version for modern Python learning is:

**Python 3**

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## 35. Python 3 Is the Modern Python

When learning Python today, you will normally use Python 3.

Example:

```
print("Hello Python 3")
```

You should avoid starting a new project with Python 2.

Python 2 is an old version and is no longer officially maintained.

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## 36. Simple Definition of Python History

**Python history** is the story of how Python was created by Guido van Rossum in the late 1980s, publicly released in 1991, and gradually developed into a powerful general-purpose programming language.

# 37. Important Terms

| Term | Meaning |
|---|---|
| Python | High-level programming language |
| Guido van Rossum | Creator of Python |
| CWI | Research institute where Python began |
| ABC | Programming language that influenced Python |
| Python 0.9.0 | First public Python release |
| Python 1.0 | First major Python release |
| Python 2.0 | Major release from 2000 |
| Python 3.0 | Major modern-generation release |
| PSF | Python Software Foundation |
| Open Source | Software whose source code is available |
| BDFL | Historical leadership title used in Python |
| Python 2 EOL | End of official Python 2 support |

# 38. Important Points to Remember

Remember these points:

1. Python was created by **Guido van Rossum**.
2. Python development began around **1989**.
3. The first public release was in **1991**.
4. The first public version was **Python 0.9.0**.
5. Python was influenced by the **ABC programming language**.
6. The name Python was inspired by **Monty Python**.
7. Python 1.0 was released in **1994**.
8. Python 2.0 was released in **2000**.
9. Python 3.0 was released in **2008**.
10. The Python Software Foundation was created in **2001**.
11. Python 2 reached end of life in **2020**.
12. Python 3 is used for modern Python development.

# 39. One-Minute Revision

```
Python
  ↓
Created by Guido van Rossum
  ↓
Development started around 1989
  ↓
First public release in 1991
  ↓
Python 1.0 → 1994
  ↓
Python 2.0 → 2000
  ↓
Python 3.0 → 2008
  ↓
Python 2 EOL → 2020
  ↓
Python 3 → Modern Python
```

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# 40. Quick Questions

### Q1. Who created Python?

**Answer:** Guido van Rossum.

### Q2. When did Python development begin?

**Answer:** Around 1989.

### Q3. When was Python first publicly released?

**Answer:** 1991.

### Q4. What was the first public Python version?

**Answer:** Python 0.9.0.

### Q5. Which programming language influenced Python?

**Answer:** ABC.

### Q6. Why is Python called Python?

**Answer:** The name was inspired by the comedy group Monty Python.

### Q7. When was Python 1.0 released?

**Answer:** 1994.

### Q8. When was Python 2.0 released?

**Answer:** 2000.

### Q9. When was Python 3.0 released?

**Answer:** 2008.

### Q10. When did Python 2 reach end of life?

**Answer:** January 1, 2020.

### Q11. Which Python version should beginners learn?

**Answer:** Python 3.

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## 41. Final Summary

Python started as a small programming project created by **Guido van Rossum**.

Development began around **1989**, and the first public version was released in **1991**.

Python continued to evolve through:

```
Python 0.9
    ↓
Python 1
    ↓
Python 2
    ↓
Python 3
```

Python 3 introduced important improvements and became the foundation for modern Python development.

Today, Python is widely used in:

- Web development
- Data science
- Artificial intelligence
- Machine learning
- Automation
- Scientific computing
- Education
- Software development

The history of Python shows how a simple programming language grew into a large and powerful programming ecosystem.

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## 42. Final Definition

> **Python is a high-level, general-purpose programming language created by Guido van Rossum. Its development began around 1989, the first public release appeared in 1991, and it has evolved through multiple generations into the modern Python 3 ecosystem.**

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